BEYOND HIGH PRICES: AUDITING INFORMATION-MEDIATED ENFORCEMENT IN RECURRENT PRICING AGENTS

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ABSTRACT

Observing high prices or correlated price movements does not prove that pricing algorithms have learned collusive punishment. We study how to separate genuine dynamic enforcement from static pricing incentives in a continuous-price duopoly where firms observe only noisy, delayed public reports and learn via recurrent proximal policy optimization (PPO). First, we establish two analytical benchmarks: a two-state public trigger model characterizing when future continuation losses deter unilateral price cuts, and a Gaussian monitoring model showing how observation noise weakens deterrence by raising both false alarms and missed detections. Second, we audit frozen GRU-PPO policies using paired counterfactual price-cut interventions and an action-independent signal ablation that serves as a proved negative control. Across ten training seeds and one million market interactions per run, normalized price elevation falls from 0.43 in early training to 0.015 under informative monitoring and 0.019 after signal ablation, with a paired difference of 0.0037 (95% bootstrap CI: $[-0.0241, 0.0288]$). Crucially, the payoff losses from forced price cuts occur almost entirely in the intervention period itself rather than through subsequent continuation losses, and several small deviations remain profitable. Rival price responses are negligible under informative monitoring and identically zero under ablation. Together, these findings distinguish apparent coordination from information-mediated enforcement and show no evidence of learned trigger punishment in this setting.

1 INTRODUCTION

High prices alone do not reveal why they persist. When pricing algorithms charge above the competitive benchmark, they may do so because exploration is incomplete, because function approximation introduces upward bias, or because unilateral price cuts trigger costly future retaliation. Only the third mechanism constitutes dynamic collusive enforcement. A credible audit of algorithmic pricing must therefore separate immediate static payoff effects from future continuation penalties rather than relying on converged price levels alone.

Our economic premise is straightforward: holding demand and cost primitives fixed, the public information channel governs whether coordinated prices can be enforced through continuation threats. In repeated games with imperfect public monitoring, the deterrent power of a trigger strategy depends on the gap between detecting a true deviation and falsely punishing cooperation, scaled by the continuation value at risk. We formalize this relationship in a fixed public trigger benchmark. Importantly, this benchmark is a conditional characterization of dynamic incentives; it does not assume that gradient-based neural learners will converge to such a strategy.

Our empirical question is correspondingly focused: do recurrent PPO pricing agents learn policies whose price levels and responses to unilateral price cuts depend on an action-sensitive public signal? To answer this, we compare an informative public reporting channel against an action-independent replacement channel, retraining both firms from scratch under identical budgets. We then freeze the trained policies and inject paired one-period price cuts to decompose total payoff changes into immediate impact effects and discounted continuation effects.

In our experiments, recurrent PPO agents do not sustain information-mediated enforcement. Average prices converge near the one-shot Nash benchmark under both informative and action-independent channels, and removing action information produces no statistically resolved drop in final prices or gain in welfare. Furthermore, counterfactual price cuts lose money immediately at impact rather than through future retaliation, while action-independent monitoring yields an identically zero impulse response by architectural invariance. Our contribution is an integrated framework combining (i) a corrected analytical benchmark for noisy public triggers, (ii) a counterfactual policy audit paired with a proved negative control, and (iii) a complete ten-seed recurrent PPO evaluation.

1.1 Research Gap: Imperfect Public Monitoring and Recurrent Policy Auditing

Classical repeated-game theory shows how firms can sustain supra-competitive prices under imperfect public monitoring when rival actions are hidden and only noisy public signals are observed (Green and Porter, 1984; Abreu et al., 1990; Fudenberg et al., 1994). In that literature, deviations are deterred when adverse public signals shift play into lower-value continuation regimes, balancing deviation detection against costly false alarms. In algorithmic pricing, Calvano et al. (2020) show that tabular Q-learning sustains supra-competitive prices under perfect price observation, and Calvano et al. (2021) extend tabular Q-learning to imperfect monitoring adapted from Green and Porter (1984), finding that discrete-price learners conditioned on short signal histories learn finite-duration "price wars" after adverse signals. Meanwhile, recent deep reinforcement learning studies such as Deng et al. (2025) compare tabular Q-learning, DQN, and PPO under perfect monitoring, focusing primarily on converged price levels.

Two gaps separate these prior studies from our work:

1. **Difference from prior imperfect public monitoring studies:** Classical theorems (Green and Porter, 1984; Abreu et al., 1990; Fudenberg et al., 1994) characterize equilibrium existence for fixed strategies in infinite-horizon games with contemporaneous signals, while Calvano et al. (2021) study discrete-grid tabular Q-learning and evaluate punishment via observational price-war trajectories. However, observing high prices or post-shock price dips does not by itself prove that future continuation losses outweigh immediate deviation gains. Unlike prior work, we audit continuous-price neural policies under delayed, two-period-averaged Gaussian reports by decomposing counterfactual price-cut payoffs into *immediate impact gain* versus *discounted continuation value change*, paired with an exact action-independent channel replacement (Proposition 1) as a proved zero-response negative control. We do not claim to refute Calvano et al. (2021) or generalize across all learners; rather, we test whether observed policy responses reflect dynamic continuation deterrence or immediate static losses.
2. **Why a windowed GRU with PPO is used:** In our market, firms observe neither rival prices nor contemporaneous one-period signals. Instead, each public report averages two consecutive latent price states, adds Gaussian noise, and arrives after a one-step queue delay. Because of this aggregation, delay, and noise—the signal-extraction problem of imperfect monitoring (Green and Porter, 1984; Calvano et al., 2021)—a single scalar report cannot distinguish a transient noise shock from a unilateral price cut. Filtering noise requires conditioning on a sequence of past reports, while continuous prices in $[1,\ 2.5]$ and continuous reports make tabular discretization over multi-period histories intractable. We therefore pair continuous-action PPO (Schulman et al., 2017) with a 128-unit GRU encoding a fixed window of 32 public reports (recomputed from a fresh hidden state at each step) as a compact finite-memory sequence encoder. We use GRU+PPO strictly as a controlled testbed for history-dependent continuous pricing policies, not as a claim that it represents commercial pricing software or outperforms tabular, MLP, or no-regret learners (see Appendix A.5 and C).

2 Market and Observation Model

Two symmetric firms $i \in \{1, 2\}$ simultaneously choose continuous prices $p_{i,t} \in [1,\ 2.5]$. Marginal cost is $c = 1$, product quality is $v = 2$, outside option utility is 0, market size is 1, and the logit scale

parameter is $\mu = 0.25$. For firm i against rival j, logit demand, stage profit, and the simulator's reported total surplus index are

$$q_i(p_i, p_j) = \frac{\exp\big((v - p_i)/\mu\big)}{1 + \exp\big((v - p_i)/\mu\big) + \exp\big((v - p_j)/\mu\big)}, \tag{1}$$

$$\pi_i(p_i, p_j) = (p_i - c)\, q_i(p_i, p_j), \tag{2}$$

$$\mathrm{TS}(p_i, p_j) = (v - c)\big[q_i(p_i, p_j) + q_j(p_i, p_j)\big]. \tag{3}$$

On a 1,501-point reference grid, the symmetric one-shot Nash price is $p^{\mathrm{N}} \approx 1.472927$ and the symmetric joint-profit-maximizing price is $p^{\mathrm{J}} \approx 1.924981$. We measure normalized price elevation as

$$E = \frac{\bar{p} - p^{\mathrm{N}}}{p^{\mathrm{J}} - p^{\mathrm{N}}},$$

which records where average price $\bar{p}$ lies relative to the Nash ($E = 0$) and joint-profit ($E = 1$) benchmarks without clipping to $[0, 1]$.

2.1 Gaussian Monitoring and Channel Replacement

At the start of each episode, a monitoring parameter vector $\theta = (\theta_{\mathrm{lo}}, \theta_{\mathrm{hi}}, \sigma_\varepsilon) \in \Theta$ is drawn uniformly from a retained MCMC calibration posterior and held fixed for the episode (full calibration details appear in Appendix B). Current prices determine the latent monitoring mean via the minimum price:

$$m_t = \theta_{\mathrm{lo}} + (\theta_{\mathrm{hi}} - \theta_{\mathrm{lo}}) \cdot \frac{\min(p_{1,t}, p_{2,t}) - p^{\mathrm{N}}}{p^{\mathrm{J}} - p^{\mathrm{N}}}, \tag{4}$$

clamped to $[\theta_{\mathrm{lo}}, \theta_{\mathrm{hi}}]$. Under the **informative channel**, the public report averages the two most recent latent means and adds a Gaussian innovation:

$$r_{t+1} = \frac{m_t + m_{t-1}}{2} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2). \tag{5}$$

Reports pass through a one-period queue delay, so r_{t+1} enters the observation window only after period-$(t+1)$ actions are chosen. Delivered reports are normalized by centering at 1.6, dividing by 0.4, and clipping to $[-5, 5]$. Under the **action-independent ablation channel**, reports are drawn directly as $r_{t+1} \sim \mathcal{N}(1.6,\, 0.4^2)$ independently of prices, keeping delay, observation dimensions, and market primitives unchanged. Because this replacement alters both action dependence and marginal report moments, comparing retrained policies identifies the effect of the full channel replacement rather than a pure information effect with matched moments (Appendix B.2).

3 Learning Architecture and Counterfactual Audit

Each firm maintains an independent actor–critic network trained via PPO (Schulman et al., 2017). At each step, the actor observes the window of the 32 most recent normalized public reports and normalized episode time $t/512$, but observes neither rival prices nor realized profits during execution. A 128-unit GRU encodes the 32-report window from a fresh zero hidden state, and its final state is concatenated with time and passed through a linear projection, $\tanh$, and layer normalization. Each run trains for 10^6 joint market interactions across 512-period episodes with discount factor $\delta = 0.99$ and no terminal value bootstrap. During training, actions are sampled from a $\tanh$-squashed Gaussian distribution; during evaluation and auditing, policies are frozen and deployed deterministically using the latent mean action $\mu(s)$ (complete hyperparameters and loss definitions are in Appendix C).

3.1 Counterfactual Price-Cut Interventions

To test whether frozen policies deter price cuts through future punishment, we run paired counterfactual rollouts. For each trained policy pair, we intervene on firm 0 or firm 1 at period $t_0 \in \{64, 256\}$ by forcing a one-period normalized action cut of 0.1 or 0.3 (corresponding to unclipped price cuts of 0.075 and 0.225). This yields $2 \times 2 \times 2 = 8$ probe types per seed, each evaluated over 128

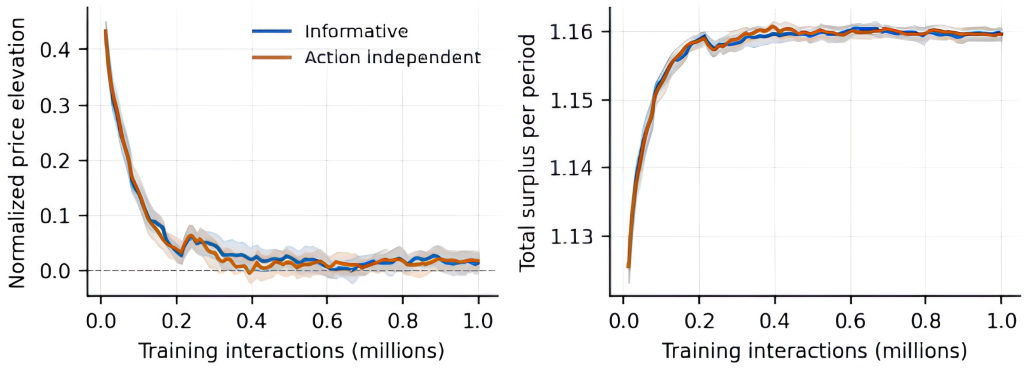


Figure 1: Deterministic deployment across training. Solid lines average ten training seeds; shaded bands show pointwise 95% confidence intervals from 2,000 whole-seed bootstrap resamples. Zero elevation corresponds to the one-shot Nash price p^{N}.

common-random-number rollouts sharing the same posterior draw θ and noise innovations ε_t. After period t_0, both firms revert to their frozen policies through period 511. We track the rival impulse response

$$\Delta p_t^{\mathrm{rival}} = p_t^{\mathrm{rival}}(\text{cut branch}) - p_t^{\mathrm{rival}}(\text{control branch})$$

and decompose the deviating firm's discounted payoff change into the immediate impact gain at t_0 and the continuation gain $\Delta\pi^{\mathrm{cont}}$ over $t > t_0$.

3.2 An Exact Negative Control

Proposition 1 (Observation invariance)**.** *Suppose execution policies are frozen deterministic functions of public report histories and time. If reports are action independent, initial buffers and innovations are coupled across branches, and actions do not enter any other actor input, then a one-period action intervention at t_0 leaves all subsequent prescribed actions unchanged. Consequently, $\Delta p_t^{\mathrm{rival}} = 0$ for all $t > t_0$ and $\Delta\pi^{\mathrm{cont}} = 0$ pathwise.*

Proof. Prior to t_0, both branches share identical inputs. At t_0, the non-intervened rival observes the same history and selects the same action. Because the report generator under ablation does not depend on prices and shares coupled innovations, the newly queued and delivered reports remain identical across branches at $t_0 + 1$. By induction, all future observation windows and prescribed actions coincide for all $t > t_0$. Since stage payoffs depend only on current prices, every post-intervention payoff difference is zero, leaving only the impact-period difference at t_0. □ □

4 Experimental Evidence Across Training Seeds

We train ten independent seed pairs from scratch under both the informative and action-independent channels. Each run logs 100 diagnostic checkpoints (16 episodes each) across training and conducts a final evaluation over 64 episodes (seed-by-seed outcomes are reported in Appendix D).

As shown in Figure 1, normalized price elevation starts near 0.43 at the first checkpoint in both conditions (0.4318 informative vs. 0.4327 action independent) as consequence of initial policy parameterization, and then declines steadily toward the Nash benchmark as learning progresses. By the final evaluation (Table 1), mean price elevation is 0.0151 under informative monitoring and 0.0188 under action-independent monitoring, with a paired difference of 0.0037 (95% bootstrap CI: $[-0.0241, 0.0288]$).

Removing action information from the public channel therefore produces no statistically resolved reduction in converged prices or improvement in surplus. Because the confidence interval spans zero with ten seeds, this result shows the absence of a large channel effect rather than formal statistical equivalence. Effective feature rank (the exponential entropy of normalized eigenvalues of centered

Table 1: Final evaluation across ten seeds (64 episodes per seed). Differences are action independent minus informative, with pointwise 95% paired bootstrap confidence intervals (2,000 resamples).

Outcome	Informative	Action independent	Paired diff. [95% CI]
Mean price	1.479804	1.481492	0.001688 $[-0.0109, 0.0130]$
Price elevation	0.015052	0.018787	0.003735 $[-0.0241, 0.0288]$
Total surplus	1.159588	1.159660	0.000072 $[-0.0013, 0.0017]$
Feature rank (firm 0)	1.274335	1.338647	0.064312 $[0.0113, 0.1337]$
Feature rank (firm 1)	1.269538	1.353578	0.084040 $[0.0282, 0.1495]$

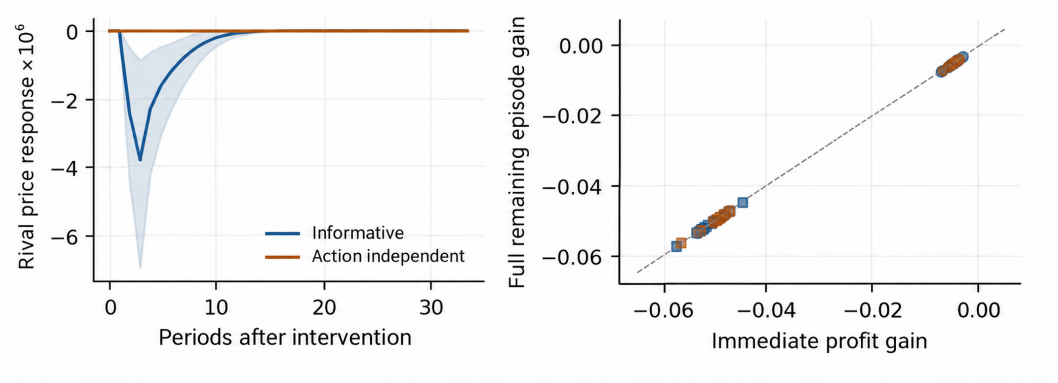


Figure 2: Left: mean rival price response $\Delta p_t^{\text{rival}}$ across eight probe types and ten seeds (in 10^{-6} price units) with 95% seed-bootstrap intervals. Right: immediate impact gain versus full remaining-episode gain for each seed, averaged by cut size (circles: 0.1 cut; squares: 0.3 cut). Points lie on the $45°$ equality line because continuation effects are negligible.

encoder features; Appendix D) is slightly higher under ablation, reflecting the change in marginal report variance rather than a causal pricing mechanism.

5 Deviation Incentives and Rival Responses

A price cut that reduces total profit is not necessarily deterred by rival retaliation; cutting prices near the Nash benchmark can simply be unprofitable immediately. Table 2 and Figure 2 therefore compare immediate impact gains against full remaining-episode gains across the two cut sizes, treating the ten training seeds as the unit of replication (all 160 individual probe estimates appear in Appendix E–F).

Under informative monitoring, the average full gain is -0.005192 for the 0.1 action cut and -0.051322 for the 0.3 action cut, while the discounted continuation component $\Delta\pi^{\text{cont}}$ averages only -5×10^{-6} in both cases. Thus, over 99.9% of the deviation loss is incurred immediately at the intervention step t_0 because baseline prices are already near p^{N}. Under action-independent monitoring, immediate and full gains are identical to machine precision ($\Delta\pi^{\text{cont}} = 0$), confirming Proposition 1.

Inspecting individual probes also reveals that not all deviations are unprofitable: 3 of 40 small-cut probes under informative monitoring (across two seeds, max gain 0.003846) and 1 of 40 under ablation (gain 0.000122) yield positive conditional mean gains. Meanwhile, the signed rival-response area under informative monitoring averages -3.7×10^{-6} price-period units (95% CI: $[-1.1 \times 10^{-5}, 3.2 \times 10^{-6}]$), which is economically negligible. We also note that because Equation (4) clamps latent means below p^{N}, cuts from near-Nash prices can saturate the signal, limiting sensitivity to shock size (Appendix B.2).

Table 2: Counterfactual price-cut audit across ten seeds. "Positive probes" counts positive conditional mean full gains among the 40 seed–firm–time probes per cut size and condition.

Channel	Action cut	Immediate gain	Full gain	Positive probes
Informative	0.1	−0.005186	−0.005192	3 of 40
Informative	0.3	−0.051317	−0.051322	0 of 40
Action independent	0.1	−0.004800	−0.004800	1 of 40
Action independent	0.3	−0.050273	−0.050273	0 of 40

6 A Fixed Public Trigger Benchmark

To clarify what dynamic enforcement would require, consider an infinite-horizon repeated stage game with discount factor $\delta \in [0, 1)$, a cooperative public state C, and an absorbing punishment state P prescribing credible stage-game Nash play (π^P). In state C, firms earn $\pi^C > \pi^P$. A bad public signal occurs with false-alarm probability $\alpha \in [0, 1)$ under cooperation and detection probability $\beta \in (0, 1]$ after a specified one-period unilateral deviation yielding current profit $\pi^D > \pi^C$.

Proposition 2 (Incentive threshold)**.** *The specified one-period deviation is unprofitable in state C if and only if*

$$\delta \;\geq\; \frac{\pi^D - \pi^C}{(\pi^D - \pi^C)(1-\alpha) + (\beta - \alpha)(\pi^C - \pi^P)}. \tag{6}$$

Equivalently, $\delta \geq \delta^*$, where

$$\delta^* = \frac{g}{g(1-\alpha) + \Delta \cdot \Delta V},$$

with $g = \pi^D - \pi^C > 0$, $\Delta = \beta - \alpha > 0$, and $\Delta V = \pi^C - \pi^P > 0$. A feasible discount factor $\delta^* < 1$ exists if and only if $\Delta \cdot \Delta V > \alpha g$.

Proof. The Bellman equations for cooperation, one-period deviation, and absorbing punishment are

$$V^C = \pi^C + \delta\big[(1-\alpha)V^C + \alpha V^P\big], \tag{7}$$

$$V^D = \pi^D + \delta\big[(1-\beta)V^C + \beta V^P\big], \qquad V^P = \pi^P + \delta V^P. \tag{8}$$

Subtracting (7) from (8) yields $V^D - V^C = g - \delta\Delta(V^C - V^P)$. Subtracting $V^P = \pi^P + \delta V^P$ from (7) gives $[1 - \delta(1-\alpha)](V^C - V^P) = \pi^C - \pi^P$, so

$$V^C - V^P = \frac{\Delta V}{1 - \delta(1-\alpha)} > 0.$$

Substituting $V^C - V^P$ into $V^D - V^C \leq 0$ gives $g[1 - \delta(1-\alpha)] \leq \delta\Delta \cdot \Delta V$. Rearranging for δ yields (6), and requiring $\delta^* < 1$ gives $\Delta \cdot \Delta V > \alpha g$ (extended derivations and numerical examples appear in Appendix A.1–A.2). □ □

Full incentive compatibility in state C requires deterring every available unilateral action a_i, not only the one maximizing current profit. With $g(a_i) = \pi_i(a_i, a_{-i}^C) - \pi^C$ and $\Delta(a_i) = \beta(a_i) - \alpha$, this requires

$$g(a_i)\big[1 - \delta(1-\alpha)\big] \;\leq\; \delta\,\Delta(a_i)\,\Delta V \quad \text{for every firm } i \text{ and action } a_i. \tag{9}$$

Because small price cuts may have low detection $\Delta(a_i)$, a finite set of probes cannot certify (9) over a continuous action space. Differentiating δ^* with $D = g(1-\alpha) + \Delta \cdot \Delta V > 0$ yields the comparative statics:

$$\frac{\partial \delta^*}{\partial \alpha} = \frac{g(g + \Delta V)}{D^2} > 0, \qquad \frac{\partial \delta^*}{\partial \beta} = -\frac{g\,\Delta V}{D^2} < 0, \qquad \frac{\partial \delta^*}{\partial (\pi^C - \pi^P)} = -\frac{g\Delta}{D^2} < 0. \tag{10}$$

Higher false alarms α erode cooperative continuation value $V^C - V^P$ and strictly raise the patience required for enforcement, while higher detection β and larger punishment gaps ΔV lower it.

7 What the Gaussian Assumption Proves

Suppose the public signal is drawn from $Y_t \sim \mathcal{N}(\mu_C, \sigma^2)$ under cooperation and $Y_t \sim \mathcal{N}(\mu_D, \sigma^2)$ after the specified deviation, and punishment is triggered when $Y_t < \tau$ for a fixed threshold $\mu_D < \tau < \mu_C$. Then

$$\alpha = \Phi\left(\frac{\tau - \mu_C}{\sigma}\right), \qquad \beta = \Phi\left(\frac{\tau - \mu_D}{\sigma}\right).$$

Proposition 3 (Restricted noise comparative static)**.** *Holding μ_C, μ_D, τ, prescribed actions, and payoff gaps fixed with $\mu_D < \tau < \mu_C$, increasing noise $\sigma > 0$ strictly decreases $\Delta = \beta - \alpha$ and strictly raises δ^* in* (6)*.*

Proof. Let $z_C = (\tau - \mu_C)/\sigma < 0$ and $z_D = (\tau - \mu_D)/\sigma > 0$. With standard normal density ϕ,

$$\frac{\partial \Delta}{\partial \sigma} = -\frac{1}{\sigma}\left[z_D\,\phi(z_D) - z_C\,\phi(z_C)\right] < 0, \tag{11}$$

since $\alpha'(\sigma) = -z_C\phi(z_C)/\sigma > 0$ and $\beta'(\sigma) = -z_D\phi(z_D)/\sigma < 0$. Combining (11) with (10) yields $\partial\delta^*/\partial\sigma = g[(g + \Delta V)\alpha'(\sigma) - \Delta V\,\beta'(\sigma)]/D^2 > 0$. As $\sigma \to 0^+$, $\alpha \to 0$ and $\beta \to 1$, so $\delta^* \to g/(g + \Delta V)$. As $\sigma \to \infty$, both $\alpha, \beta \to 1/2$, so $\delta^* \to 2$, which is infeasible for $\delta < 1$ (Appendix A.3). □ □

7.1 Conditional Connection to the PPO Environment

Propositions 2 and 3 provide an exact incentive accounting benchmark, but three structural differences separate this analytical model from the PPO experiment: (i) the PPO simulator delivers continuous, two-period-averaged, one-step-delayed reports to a 32-step GRU rather than enforcing a contemporaneous binary threshold; (ii) the experiment compares an informative channel against an action-independent replacement rather than sweeping σ across a fixed trigger policy; and (iii) PPO episodes terminate at $T = 512$ with observable normalized time rather than running over an infinite horizon. Consequently, our empirical audit tests whether trained recurrent policies exhibit continuation-value deterrence in practice, without claiming that finite-horizon PPO must replicate infinite-horizon trigger equilibria.

8 Interpretation, Limitations, and Conclusion

Imperfect public monitoring theory shows that sustaining collusion requires linking noisy public signals to continuation value losses (Green and Porter, 1984; Abreu et al., 1990; Fudenberg et al., 1994). While tabular Q-learning can learn supra-competitive pricing under both perfect monitoring (Calvano et al., 2020) and imperfect monitoring with finite price wars (Calvano et al., 2021), and deep RL studies report near-Nash pricing under perfect-monitoring PPO (Deng et al., 2025), our contribution lies in auditing *how* deviations are penalized via counterfactual payoff decomposition and channel replacement.

Five methodological limitations bound our conclusions (expanded in Appendix G–H):

- **Input distribution confound:** Replacing the informative channel with $r_{t+1} \sim \mathcal{N}(1.6, 0.4^2)$ alters marginal report moments alongside action dependence; isolating a pure information effect requires moment-matched placebo channels.
- **Signal saturation and probe scope:** Clamping in (4) saturates latent means at or below p^{N}, and our audit tests a finite set of one-step cuts rather than best-response or multi-step deviations.
- **Missing positive control:** Proposition 1 validates our zero-response negative control, but evaluating audit sensitivity (statistical power) requires testing against a known collusive trigger positive control.
- **Finite horizon and model scope:** Known 512-period termination with time inputs and ten training seeds in a single stationary market bound generalizability; an interval spanning zero does not prove exact treatment equivalence.

- **Reproducibility accounting:** All 20 PPO runs (58.17M total interactions; 44.06 hours; MCMC split $\hat{R} \leq 1.0089$, ESS ≥ 606.86 (Vehtari et al., 2021)) are verified by SHA256 hashes in Appendix G.

Conclusion. High prices and post-shock price movements must not be conflated with dynamic collusive enforcement. By combining a corrected noisy-trigger benchmark with a counterfactual price-cut audit and an exact negative control, we show that in this recurrent PPO setting, early high prices decay to near-Nash levels and deviation losses are overwhelmingly immediate rather than continuation-driven. Prices are directly observable; information-mediated enforcement remains a structural hypothesis that must be audited directly.

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AI USE DISCLOSURE

Generative AI was used to assist with manuscript drafting and revision, mathematical derivation checks, implementation review, postprocessing of archived experimental outputs, and figure preparation. All numerical results were computed from the archived simulator runs rather than generated hypothetically, and the authors retain full responsibility for all assumptions, derivations, code, references, and claims.